

List

Part-2



Agenda

- ① Packing and Unpacking
- ② built-in methods
- ③ list() method
- ④ comparison operator on list
- ⑤ concatenation operator
- ⑥ Repetition operator
- ⑦ Slicing operator

Packing and Unpacking

$l1 = [20, 50, 30]$

$a, b, c = l1$ $\neq$ unpacking

number of variables in the left hand side must be same as the number of elements in the list.

$a = 5$

$b = 6$

$c = 10$

$l2 = [a, b, c]$

$\neq$ packing

Built in methods

- `len()` → returns length of specified iterable
- `min()` → returns min value element
- `max()` → returns max value element
- `sum()` → returns sum of elements
- `sorted()` → returns a sorted list of elements.

`l = [24, 13, 5]`

`l1 = sorted(l)`

`[5, 13, 24]`

list() method

`l1 = list()` $\neq$ `l1 = []`

`l1 = list(10)` $\neq$ ~~`l1 = [10]`~~ Error

`l1 = list(10, 20, 30)`

`l1 = list("mySirG")`

`l1 = list(range(5))`

`l1 = list([10, 20])`

list() method can take at most
one argument.

one argument $\longrightarrow$ iterable

Comparison Operator on list

$l1 = [1, 2, 3]$

$l2 = [2, 3, 1]$

$l3 = [1, 2, 3, 4, 5]$

$l4 = [1, 2, 3]$

$l1 == l2$ False

$l1 == l3$ False

$l1 == l4$ True

$l1 > l2$ False

Concatenation Operator

$$l1 = [1, 5, 9]$$

0	1	2
1	5	9

$$l2 = [2, 3, 1]$$

0	1	2
2	3	1

$$l3 = l1 + l2$$

0	1	2	3	4	5
1	5	9	2	3	1

$$l1 += l2 \rightarrow l1 = l1 + l2$$

Repetition Operator

$l1 = [2, 5]$

0	1
2	5

$l1 * 5$

0	1	2	3	4	5	6	7	8	9
2	5	2	5	2	5	2	5	2	5

Slicing Operator

listObject [beg : end : step]

l1 = [20, 40, 10, 30, 60, 50]

By default Step = 1

By default beg, end = extreme end

l1[2:6:2]

l1[4:0:-1]